Oriented Processes

Group-valued data, a chaos constant, and an equivariant COMM
for the reflective higher-order calculus

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Abstract

We take the pattern-matching variant of the rho calculus parametric in a type of data, and specialise the data to a group G. The group acts on processes freely: an element g multiplies the datum of every name, descending recursively through quotation, and the multiplication is deferred at variables until they are instantiated. To the calculus we add an annihilating constant $\mathbf{1}$, read as chaos or divergence, with $\mathbf{1} \mid P \equiv \mathbf{1}$, and an idempotent summation. The communication rule is generalised so that a receive and a send on two orientations of the same channel produce a sum of the continuation transported by the relative orientation and by its inverse. We show that the action is an automorphism of the calculus, that chaos is forced to have every name free, and that reduction is G-equivariant for an arbitrary group exactly when the two coefficients are chosen as $\nu(x)\nu(u)^{-1}$ and $\nu(u)\nu(x)^{-1}$. Because the action is deferred, substitution and matching commute with it outright, and the action is free on open terms as well as closed ones. It then yields an orbit-counting theorem: the symmetry group of a process with finitely many free atoms acts freely on them, so its order divides their number. For rotation groups this is a crystallographic restriction on processes. Three applications indicate the utility of the structure: symmetry reduction for state-space exploration, where the symmetry is read off the term rather than declared; symmetric rings, including the dining philosophers, as orbit products; and *twisted roots* of processes, which generalise the n-th roots of radical fault tolerance and replace its divisibility obstruction by a symmetry condition in which the stabiliser of a component plays the part of its replication factor.

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1 Introduction

Earlier work treated the syntax of the rho calculus as a type constructor parametric in a type D of data carried by names and messages, and settled on a pattern-matching binder that lets a receiver bind every component of what it receives, so that the datum on a message is no longer observationally inert [9]. That construction asks nothing of D. This note asks what the calculus gains when D is a group.

The answer begins with an action. A group acts on itself by left multiplication, and that action lifts to processes by multiplying the datum of every name a process mentions, at every depth of quotation. The lifted action is free on names: no non-identity element fixes a name. Freeness turns questions about the symmetry of a process into counting arguments, and the calculus acquires a notion of *orientation*. The same channel code may occur at many orientations, and a rotation of a process moves all of them at once.

Two further ingredients complete the setting. The first is a constant $\mathbf{1}$ for chaos: the process that may do anything, so that composing anything with it adds nothing. The second is a summation, which the communication rule needs because its reduct is a sum. The rule itself is the point of the construction. A receive on one orientation of a channel and a send on another communicate, and the continuation is delivered twice, once transported by the relative orientation of the two ends and once by its inverse. When the two ends agree, the transports cancel and the ordinary rho communication is recovered.

Contributions.

- The calculus ρ_G (Section 2): syntax, the free action, chaos, idempotent summation, matching, and the oriented COMM rule.

- Basic observations (Section 3): the action is an automorphism; chaos has every name free; substitution commutes with the action and matching is invariant under it; reduction is G-equivariant (Theorem 3.5), and so are context-labelled transitions; the calculus is conservative over the original; and an orbit-counting theorem for open and closed terms (Theorem 3.10) with a crystallographic corollary for rotation groups (Corollary 3.13).
- Three applications: symmetry reduction with symmetry computed from the term (Section 4); symmetric rings as orbit products (Section 5); and twisted roots of processes, with a divisibility theorem that interpolates between replication and sharding (Section 6). The unique decomposition it needs holds up to structural congruence and is proved, not assumed (Lemma 6.2).

Scope. Summation is taken idempotent throughout, and chaos absorbs it. Other readings of summation are possible and are not pursued here.

2 The calculus

2.1 Syntax

Fix a group $(G, \cdot, e, (-)^{-1})$ and a set V of variables, divided into process variables $X, Y, \ldots$ and data variables $\delta, \delta', \ldots$. We write e for the identity of G, reserving $\mathbf{1}$ for chaos. The terms of ρ_G are

$$\begin{array}{rcll} d & ::= & g \mid [g]\delta & g \in G \quad \text{(data)} \\ x, u, \pi & ::= & @\langle P, d\rangle & \text{(names; patterns)} \\ P, Q & ::= & \mathbf{0} \mid \mathbf{1} \mid [g]X \mid \mathtt{for}(\pi \leftarrow x)P \mid x!(\langle Q, d\rangle) \mid P \mid Q \mid P + Q & \text{(processes)} \end{array}$$

Variables occur *decorated*: $[g]X$ and $[g]\delta$ carry a group element, the action that is pending on the variable until it is instantiated. We write X for $[e]X$ and δ for $[e]\delta$. A name $@\langle P, d\rangle$ pairs a *code* P with a datum d; we write $\mathrm{code}(@\langle P, d\rangle) = P$ and $\nu(@\langle P, d\rangle) = d$. A message $x!(\langle Q, d\rangle)$ ships the pair $\langle Q, d\rangle$ on x; at communication it is received as the name $@\langle Q, d\rangle$.

In a receive $\mathtt{for}(\pi \leftarrow x)P$ the pattern π is a name in which every variable occurring is bound by the receive, and each occurs exactly once, possibly decorated. The canonical pattern $@\langle X, \delta\rangle$ binds both the code and the datum of what is received. As in [9], process variables play the part the drop operator plays in the original calculus [7]: a received code is used by writing its variable in process position. A name is *closed* when it contains no variables, and a process is closed when every variable in it is bound. Reduction is defined on closed processes.

Remark 2.1 (Notation). The sketch that motivated this note wrote the action with $*$. Since $*$ is the drop operator in the rho calculus, we write the action as $g \cdot P$. For the same reason the identity of G is e, leaving $\mathbf{1}$ free for chaos.

2.2 The free action

Definition 2.2 (Action)**.** For $g \in G$ define $g \cdot -$ on data, names and processes by

$$\begin{array}{lll} g \cdot h = gh \quad (h \in G) & g \cdot [k]\delta = [gk]\delta & g \cdot [k]X = [gk]X \\ g \cdot \mathbf{0} = \mathbf{0} & g \cdot \mathbf{1} = \mathbf{1} & g \cdot @\langle P, d\rangle = @\langle g \cdot P, g \cdot d\rangle \end{array}$$

$$g \cdot (P \mid Q) = g \cdot P \mid g \cdot Q \qquad g \cdot (P + Q) = g \cdot P + g \cdot Q$$
$$g \cdot \texttt{for}(\pi \leftarrow x)P = \texttt{for}(g \cdot \pi \leftarrow g \cdot x)g \cdot P \qquad g \cdot x!(\langle Q, d\rangle) = g \cdot x!(\langle g \cdot Q, g \cdot d\rangle)$$

The action multiplies every literal datum on the left, at every depth of quotation, inside patterns as well as in names and messages. At a variable it is not applied but recorded, and pending actions compose, $h \cdot [k]X = [hk]X$, so each occurrence carries a single element. The record is discharged by substitution:

$$([g]X)\sigma = g \cdot \sigma(X), \qquad ([g]\delta)\sigma = g\,\sigma(\delta).$$

The action renames nothing, so it needs no side condition against capture. On closed terms it is the image of left multiplication $L_g : G \to G$ under the functor that the parametric syntax defines.

Local frames. A decoration has a reading. A bound variable holds what was received in the receiver's own frame, and its decoration records the transformation from that frame to the frame in which the body is written. A rotated copy of a receive therefore binds exactly what the unrotated one binds, and the rotation is applied when the variable is used. Changing the frame of a bound variable uniformly, at its binding occurrence and in the body, changes nothing observable; this is the axiom (FRAME) below.

2.3 Structural congruence

Structural congruence $\equiv$ is the least congruence, closed under every term former including quotation, that satisfies the axioms of Table 1. Two names are equal when their codes are structurally congruent and their data are equal.

(PAR)	$P \mid \mathbf{0} \equiv P,\ P \mid Q \equiv Q \mid P,\ (P \mid Q) \mid R \equiv P \mid (Q \mid R)$	monoid
(SUM)	$P + Q \equiv Q + P,\ (P + Q) + R \equiv P + (Q + R),\ P + P \equiv P$	semilattice
(DIST)	$P \mid (Q + R) \equiv P \mid Q + P \mid R$	distributivity
(CHAOS)	$\mathbf{1} \mid P \equiv \mathbf{1},\ \mathbf{1} + P \equiv \mathbf{1}$	chaos absorbs both
(FRAME)	$\texttt{for}(\pi \leftarrow x)P \equiv \texttt{for}(\pi^{Z,h} \leftarrow x)P^{Z,h}$	change of frame

Table 1: Axioms of structural congruence.

In (FRAME), Z is a variable bound by π, $h \in G$, and $T^{Z,h}$ replaces every occurrence $[k]Z$ in T by $[kh]Z$: right multiplication of the decorations of one bound variable. It is a frame α-conversion. A matcher for $\pi^{Z,h}$ binds Z to h^{-1} times what a matcher for π binds, and the body occurrences, multiplied by h on the right, discharge to the same result. Every receive is congruent to one whose binding occurrences are undecorated, so closed terms are identified exactly as they would be if the action left variables alone.

The axiom $\mathbf{1} \mid P \equiv \mathbf{1}$ is the defining property of chaos: it represents all possible behaviours, so composing it with any P cannot enrich it. The same reasoning motivates $\mathbf{1} + P \equiv \mathbf{1}$. The two are consistent with (DIST): $(\mathbf{1} + P) \mid Q \equiv \mathbf{1} \mid Q + P \mid Q \equiv \mathbf{1}$. Note that $\mathbf{0}$ is not a unit for $+$; it is the inert configuration, one alternative among others.

By (DIST), every process is congruent to a finite sum of sum-free *configurations*, or to $\mathbf{1}$. Summation is therefore an ensemble of alternatives, and reduction (below) refines one member of the ensemble.

2.4 Matching

Matching is performed on normal forms. Orient (CHAOS) and (DIST) from left to right, remove duplicate summands, and use (FRAME) to make every binding occurrence undecorated; write $\mathrm{nf}(P)$ for a normal form of P modulo associativity and commutativity of $\mid$ and $+$ (AC). A substitution σ maps process variables to closed processes and data variables to elements of G, and acts on decorated occurrences as above.

Definition 2.3 (Matching)**.** For a pattern π and a closed name n,

$$\mathrm{match}(\pi, n) \;=\; \{\, \sigma \;:\; \pi\sigma =_{AC} \mathrm{nf}(n) \,\}.$$

Matching modulo AC can have several solutions, and a communication may use any of them. A decorated pattern variable is solved by inversion: $[g]X$ matches Q exactly with $\sigma(X) = g^{-1} \cdot Q$, and $[g]\delta$ matches d with $\sigma(\delta) = g^{-1}d$. This is where the group axioms enter the definition of communication itself: under a monoid action a pending decoration could leave a pattern with no solution or with many.

One further convention in the definition is deliberate.

Remark 2.4 (Chaos in quotes). Matching is *not* performed modulo (CHAOS). Otherwise the pattern $@\langle X \mid Y, \delta\rangle$ would match $@\langle \mathbf{1}, d\rangle$ under every substitution $\{\mathbf{1}/X, Q/Y\}$, since $\mathbf{1} \mid Q \equiv \mathbf{1}$ for all Q, and matching against quoted chaos would not be finitary. In normal form chaos is an atom: it matches a bare process variable and nothing composite.

2.5 Reduction

The *relative orientation* of a receive on x and a send on u is $\nu(x)\nu(u)^{-1}$. The reduction relation $\rightsquigarrow$ on closed processes is the least relation closed under the rules of Table 2.

(COMM)	$\dfrac{\mathrm{code}(x) \equiv \mathrm{code}(u) \qquad \sigma \in \mathrm{match}(\pi, @\langle Q, d\rangle)}{\mathtt{for}(\pi \leftarrow x)P \mid u!(\langle Q, d\rangle) \;\rightsquigarrow\; (\nu(x)\nu(u)^{-1}) \cdot P\sigma \;+\; (\nu(u)\nu(x)^{-1}) \cdot P\sigma}$
(PAR)	$\dfrac{P \rightsquigarrow P'}{P \mid R \rightsquigarrow P' \mid R}$
(SUM)	$\dfrac{P \rightsquigarrow P'}{P + R \rightsquigarrow P' + R}$
(STRUCT)	$\dfrac{P \equiv P' \qquad P' \rightsquigarrow Q' \qquad Q' \equiv Q}{P \rightsquigarrow Q}$
(DIV)	$\mathbf{1} \rightsquigarrow \mathbf{1}$

Table 2: Reduction rules.

The premise $\mathrm{code}(x) \equiv \mathrm{code}(u)$ is the *gate* of the rule. A receive and a send interact when they are on the same channel code, possibly at different orientations, and the relative orientation then acts on the continuation. The two coefficients are mutually inverse, so writing $h = \nu(x)\nu(u)^{-1}$ the reduct is $h \cdot P\sigma + h^{-1} \cdot P\sigma$.

Remark 2.5 (The gate). Other gates are possible.

- The *strict* gate $x = u$ makes every coefficient e, and the calculus collapses to the original one extended by chaos and summation.
- The *open* gate, always true, lets every receive meet every send whose message matches its pattern. Pattern matching is then the only selector of partners, and channel discipline is lost.

- A *K-gate*, $\mathrm{code}(x) \equiv \mathrm{code}(u)$ and $\nu(x)\nu(u)^{-1} \in K$ for a subset $K \subseteq G$ that contains e, is closed under inverses and is closed under conjugation, restricts interaction between orientations of a channel to the Cayley graph of G with connection set K.

All of the results below hold for each of these gates; we fix the coded gate as the default because it is the least change that lets orientation matter.

Remark 2.6 (Order of the second coefficient). The sketch that motivated this note used $\nu(x)^{-1}\nu(u)$ as the second coefficient. In an abelian group it coincides with $\nu(u)\nu(x)^{-1}$. In a non-abelian group it is the $\nu(x)$-conjugate of the inverse of the first coefficient, and with that choice reduction is equivariant only for elements commuting with $\nu(x)^{-1}\nu(u)$; see Remark 3.6. The choice $\nu(u)\nu(x)^{-1}$ is the one that makes reduction equivariant for every group, and in particular for non-abelian rotation groups such as $SO(3)$.

3 Basic observations

3.1 The action is an automorphism

Lemma 3.1. *(i)* $e \cdot P = P$ *and* $g \cdot (h \cdot P) = (gh) \cdot P$.

(ii) $P \equiv Q$ *if and only if* $g \cdot P \equiv g \cdot Q$.

(iii) $\mathrm{nf}(g \cdot P) =_{AC} g \cdot \mathrm{nf}(P)$.

So each $g \cdot -$ *is an automorphism of* $(\mathcal{P}/\equiv, |, +, \mathbf{0}, \mathbf{1})$*, and* G *acts on processes up to structural congruence.*

Proof. (i) is by induction on terms; the base cases are a literal datum and a decorated variable, where $g(hk) = (gh)k$. For (ii), the action maps every instance of an axiom in Table 1 to an instance of the same axiom, because it commutes with the term formers and fixes $\mathbf{0}$ and $\mathbf{1}$; for (FRAME) this is the commutation of left and right multiplication. Hence it preserves $\equiv$, and $g^{-1} \cdot -$ is its inverse by (i). (iii) holds because the oriented normalisation rules are themselves invariant under the action. □

3.2 Chaos has every name

Call a name occurrence in P *free* when every variable in it is free in P; for closed P these are the closed names. The *free atoms* $\mathrm{FA}(P)$ of P are its free names in subject position or in a payload, together with its free variable occurrences $[k]X$ in process position:

$$\mathrm{FA}(\mathbf{0}) = \varnothing, \qquad \mathrm{FA}([k]X) = \{[k]X\},$$
$$\mathrm{FA}(P \mid Q) = \mathrm{FA}(P) \cup \mathrm{FA}(Q), \qquad \mathrm{FA}(P + Q) = \mathrm{FA}(P) \cup \mathrm{FA}(Q),$$
$$\mathrm{FA}(x!(\langle Q, d\rangle)) = \{x\}^\circ \cup \mathrm{FA}(Q), \qquad \mathrm{FA}(\mathtt{for}(\pi \leftarrow x)P) = \{x\}^\circ \cup (\mathrm{FA}(P) \setminus \mathrm{bound}(\pi)),$$

where $\{x\}^\circ$ is $\{x\}$ if the occurrence is free and empty otherwise, and $\mathrm{FA}(P) \setminus \mathrm{bound}(\pi)$ discards atoms mentioning a variable bound by π. Write $\mathrm{FN}(P)$ for the names among the free atoms, and $\mathcal{A}$ for the set of all atoms. $\mathrm{FA}(\mathbf{1})$ is fixed by the following proposition.

Proposition 3.2. *If* FA *respects* $\equiv$ *and is a union over* $|$*, then* $\mathrm{FA}(\mathbf{1}) = \mathcal{A}$*: chaos has every name, and every variable, free.*

Proof. For any name x, $\mathrm{FA}(\mathbf{1}) = \mathrm{FA}(\mathbf{1} \mid x!(\langle \mathbf{0}, e\rangle)) = \mathrm{FA}(\mathbf{1}) \cup \{x\}$, so $x \in \mathrm{FA}(\mathbf{1})$; likewise with $[k]X$ in place of the send. □

Chaos has every name free because it is absorbing. Two consequences are worth recording at once. First, $g \cdot \mathbf{1} = \mathbf{1}$ is no longer an axiom about the action but an instance of its definition, since left multiplication permutes $\mathcal{A}$. Second, no name is fresh for chaos: an argument that picks a channel not free in P fails as soon as $P \equiv \mathbf{1}$.

Lemma 3.3. $\mathrm{FA}(g \cdot P) = g \cdot \mathrm{FA}(P)$, *and* FA *respects* $\equiv$.

Proof. The first claim is by induction: the action preserves freeness of occurrences and permutes $\mathcal{A}$. The second is a check of the axioms. For (CHAOS), $\mathrm{FA}(\mathbf{1} \mid P) = \mathcal{A} \cup \mathrm{FA}(P) = \mathcal{A} = \mathrm{FA}(\mathbf{1})$; (FRAME) changes only bound variables, which free atoms do not mention. □

Without deferral this would fail in an instructive way. If the action left variables alone, a bare variable X and a name $@\langle X, \delta\rangle$ built only from variables would be fixed by the whole group, and freeness would hold only for closed names. Deferral makes every atom move.

3.3 Substitution and matching

Lemma 3.4. $g \cdot (P\sigma) = (g \cdot P)\sigma$, *and* $\mathrm{match}(g \cdot \pi, g \cdot n) = \mathrm{match}(\pi, n)$.

Proof. The substitution identity is by induction on P. At a decorated process variable, $(g \cdot [k]X)\sigma = ([gk]X)\sigma = (gk) \cdot \sigma(X) = g \cdot ([k]X)\sigma$ by Lemma 3.1(i), and a decorated data variable is the same calculation in G. At a literal datum both sides give gh. For matching, $\pi\sigma =_{AC} \mathrm{nf}(n)$ if and only if $g \cdot (\pi\sigma) =_{AC} g \cdot \mathrm{nf}(n)$, because the action is a bijection that respects AC; by the substitution identity and Lemma 3.1(iii) this reads $(g \cdot \pi)\sigma =_{AC} \mathrm{nf}(g \cdot n)$. □

The action passes through substitution without touching it, and a rotated redex is matched by the *same* substitution as the original. For example, $@\langle X, h\rangle$ matches $@\langle Q, d\rangle$ exactly when $h = d$, with $X \mapsto Q$; after the action it is $@\langle [g]X, gh\rangle$, which matches $@\langle g \cdot Q, gd\rangle$ exactly when $gh = gd$, again with $X \mapsto Q$. Acting on patterns is what keeps a rotated redex a redex, and deferral is what keeps its matcher fixed.

3.4 Equivariance

Theorem 3.5 (Equivariance)**.** *If $P \rightsquigarrow P'$ then $g \cdot P \rightsquigarrow g \cdot P'$, for every $g \in G$.*

Proof. By induction on the derivation. (PAR), (SUM) and (STRUCT) follow from Lemma 3.1, and (DIV) from $g \cdot \mathbf{1} = \mathbf{1}$. For (COMM), let $a = \nu(x)$ and $b = \nu(u)$. The rotated redex is $\mathtt{for}(g \cdot \pi \leftarrow g \cdot x)g \cdot P \mid g \cdot u!(\langle g \cdot Q, gd\rangle)$. Its gate holds because $\mathrm{code}(g \cdot x) = g \cdot \mathrm{code}(x)$ and Lemma 3.1(ii). By Lemma 3.4, σ itself is a matcher of the rotated redex, and the continuation becomes $(g \cdot P)\sigma = g \cdot (P\sigma)$. The rotated data are ga and gb, so the coefficients are

$$(ga)(gb)^{-1} = g\,ab^{-1}g^{-1}, \qquad (gb)(ga)^{-1} = g\,ba^{-1}g^{-1}.$$

By Lemma 3.1(i) the reduct is

$$(g\,ab^{-1}g^{-1}) \cdot g \cdot (P\sigma) + (g\,ba^{-1}g^{-1}) \cdot g \cdot (P\sigma) = g \cdot \big((ab^{-1}) \cdot P\sigma + (ba^{-1}) \cdot P\sigma\big),$$

which is g applied to the original reduct. □

Remark 3.6 (The non-abelian case). With $a^{-1}b$ as the second coefficient, the rotated coefficient is $(ga)^{-1}(gb) = a^{-1}b$, unchanged, and the second summand becomes $(a^{-1}b\,g) \cdot P\sigma$ where equivariance needs $(g\,a^{-1}b) \cdot P\sigma$. These agree for all P only when g commutes with $a^{-1}b$. Equivariance is therefore what selects the coefficients.

Contexts and labels. A context C is a term with one hole, which may sit under receives that bind variables of what fills it. Let $g \cdot C$ act on everything but the hole. By induction on C,

$$(g \cdot C)[g \cdot P] \;=\; g \cdot (C[P]),$$

for open P as well as closed: a variable of P captured by a pattern of C is decorated on both sides by the same element. Write $P \xrightarrow{C} P'$ when $C[P] \rightsquigarrow P'$, the transitions of context-labelled bisimulation.

Corollary 3.7. *$P \xrightarrow{C} P'$ if and only if $g \cdot P \xrightarrow{g \cdot C} g \cdot P'$. Consequently, if a set of observing contexts $\mathcal{O}$ is closed under the action, then the largest $\mathcal{O}$-bisimulation $\approx_{\mathcal{O}}$ is equivariant: $P \approx_{\mathcal{O}} Q$ if and only if $g \cdot P \approx_{\mathcal{O}} g \cdot Q$.*

Proof. The first claim is Theorem 3.5 applied to $C[P]$, with g^{-1} for the converse. For the second, if $\mathcal{R}$ is an $\mathcal{O}$-bisimulation then so is $\{(g \cdot P, g \cdot Q) : (P, Q) \in \mathcal{R}\}$, by the first claim and closure of $\mathcal{O}$. □

3.5 Conservativity and frames

Proposition 3.8 (Conservativity)**.** *If $\nu(x) = \nu(u)$, in particular if $x = u$, then $\mathtt{for}(\pi \leftarrow x)P \mid u!(\langle Q, d\rangle) \rightsquigarrow P\sigma$. Hence for the trivial group, and on any process whose communications occur between equal names, ρ_G reduces exactly as the pattern-matching calculus of [9], extended by chaos and summation.*

Proof. Both coefficients are e, so the reduct is $P\sigma + P\sigma \equiv P\sigma$ by idempotence. □

This is the cancellation in the rule. The relative orientation of matched ends is the identity, and the doubling of the continuation disappears into the idempotence of summation.

Symmetry between the ends. Exchanging the roles of $\nu(x)$ and $\nu(u)$ exchanges the two summands of the reduct, which is unchanged. The rule does not depend on whose frame the relative orientation is measured in.

Frames. Consider a name n in the receiver's frame, $\nu(n) = \nu(x)$. The second summand moves it to $\nu(u)\nu(x)^{-1}\nu(x) = \nu(u)$: into the sender's frame. The first moves it to $\nu(x)\nu(u)^{-1}\nu(x)$, the reflection of the sender's frame through the receiver's; in an abelian group written additively, a receiver at j hearing from k lands at k in one summand and at $2j - k$ in the other. The bound variables of the continuation move with it: what was received is held in the receiver's frame, and the coefficient is applied to it when it is used, by Lemma 3.4.

Anisotropic and isotropic codes. A code c is *isotropic* for $H \leq G$ when $h \cdot c \equiv c$ for all $h \in H$; nameless codes such as $\mathbf{0}$ are isotropic for all of G. All orientations $@\langle c, h\rangle$ of an isotropic code share a code, so under the coded gate they communicate with one another, with non-trivial coefficients. A code with names is *anisotropic*: rotating a name moves its code as well as its datum, and by Theorem 3.10 below only finitely many rotations can fix it. Anisotropic codes therefore produce genuinely distinct channels under rotation, and processes built from them reduce exactly as in the original calculus (Proposition 3.8). Rotation can thus be used either to *place* copies of a process on distinct channels, or to *orient* a single channel.

3.6 Stabilisers and orbit counting

Definition 3.9. The *symmetry group* of P is $\mathrm{Stab}(P) = \{\, g \in G : g \cdot P \equiv P \,\}$.

By Lemma 3.1, $\mathrm{Stab}(P)$ is a subgroup, $\mathrm{Stab}(g \cdot P) = g\,\mathrm{Stab}(P)\,g^{-1}$, $\mathrm{Stab}(P \mid Q) \supseteq \mathrm{Stab}(P) \cap \mathrm{Stab}(Q)$, and $\mathrm{Stab}(\mathbf{0}) = \mathrm{Stab}(\mathbf{1}) = G$.

Every atom has a *frame* ϕ: for a name, $\phi(@\langle Q, h\rangle) = h$ and $\phi(@\langle Q, [k]\delta\rangle) = k$; for a variable, $\phi([k]X) = k$. The action multiplies frames, $\phi(g \cdot a) = g\,\phi(a)$.

Theorem 3.10 (Orbit counting)**.** $\mathrm{Stab}(P)$ *acts freely on* $\mathrm{FA}(P)$*. Consequently, if* $\mathrm{FA}(P)$ *is finite, then either* $\mathrm{FA}(P) = \emptyset$ *and* $\mathrm{Stab}(P) = G$*, or* $\mathrm{Stab}(P)$ *is finite and* $|\mathrm{Stab}(P)|$ *divides* $|\mathrm{FA}(P)|$*.*

Proof. If $g \in \mathrm{Stab}(P)$, Lemma 3.3 gives $g \cdot \mathrm{FA}(P) = \mathrm{FA}(g \cdot P) = \mathrm{FA}(P)$, so $\mathrm{Stab}(P)$ acts on $\mathrm{FA}(P)$. The action is free: $g \cdot a = a$ requires $g\,\phi(a) = \phi(a)$, hence $g = e$. If $\mathrm{FA}(P)$ is finite and non-empty, every orbit of $\mathrm{Stab}(P)$ on it has $|\mathrm{Stab}(P)|$ elements, so $\mathrm{Stab}(P)$ is finite and its order divides $|\mathrm{FA}(P)|$. If $\mathrm{FA}(P) = \emptyset$, then P contains no receive, no send, no free variable and no chaos outside quotation, so it is built from $\mathbf{0}$ and the two operators and every g fixes it. □

$\mathrm{FA}(P)$ is finite exactly when $\mathbf{1}$ does not occur in P outside quotation, and the theorem applies to open terms as well as closed ones. So it covers every process that is not chaotic, and chaos is the one way for a process with atoms to be fixed by an infinite group. For closed P the free atoms are the free names.

Corollary 3.11. *If* G *is infinite and* $\mathrm{Stab}(P) = G$*, then* $\mathrm{FA}(P) = \emptyset$ *or* $\mathrm{FA}(P) = \mathcal{A}$*. In words: the only processes invariant under an infinite group are those that mention no names or free variables, and those that contain chaos.*

Proposition 3.12 (Symmetry is computable)**.** *If* $\mathrm{FA}(P)$ *is finite and non-empty, fix* $a_0 \in \mathrm{FA}(P)$*. Then*

$$\mathrm{Stab}(P) \subseteq \{\, \phi(a)\,\phi(a_0)^{-1} : a \in \mathrm{FA}(P) \,\},$$

so $\mathrm{Stab}(P)$ *is found by at most* $|\mathrm{FA}(P)|$ *congruence checks, for any group, finite or not.*

Proof. If $g \in \mathrm{Stab}(P)$ then $g \cdot a_0 \in \mathrm{FA}(P)$, and $\phi(g \cdot a_0) = g\,\phi(a_0)$ determines g. □

3.7 Rotations

Take the data to be orientations. With $G = SO(2)$ a name is a channel code at a planar orientation, and with $G = SO(3)$ at an orientation in space.

Corollary 3.13 (Crystallographic restriction)**.** *Let* $\mathrm{FA}(P)$ *be finite and non-empty.*

(i) If $G = SO(2)$, then $\mathrm{Stab}(P)$ is cyclic of order dividing $|\mathrm{FA}(P)|$.

(ii) If $G = SO(3)$, then $\mathrm{Stab}(P)$ is cyclic, dihedral, tetrahedral, octahedral or icosahedral, of order dividing $|\mathrm{FA}(P)|$.

Proof. By Theorem 3.10 the stabiliser is a finite subgroup, and these are the finite subgroups of $SO(2)$ and $SO(3)$ [1]. □

A process can have the rotational symmetry of a polygon or a polyhedron, but never continuous rotational symmetry unless it mentions no names or is chaotic. An icosahedrally symmetric process mentions a multiple of 60 names.

3.8 Orbit products and orbit sums

For a finite subgroup $H \leq G$ and a process R define

$$\prod_H R \;=\; \big|_{h\in H}\, h \cdot R, \qquad \sum_H R \;=\; \sum_{h\in H}\; h \cdot R.$$

Both are H-invariant: $k \cdot \prod_H R$ re-indexes the product by $h \mapsto kh$, and associativity and commutativity of $|$ do the rest; the same holds for $\sum_H R$. Orbit products are the standard way to impose a symmetry on a process, with R playing the part of a fundamental domain. The reduct of (COMM) is a partial orbit sum, over the two-element set $\{h, h^{-1}\}$, and is invariant under the inversion that exchanges them.

4 Application I: symmetry reduction

Explicit-state exploration of concurrent systems is limited by the size of the state space, and symmetry reduction is one of the standard remedies [4, 5, 3]: when a group acts on the states by automorphisms of the transition relation, it suffices to explore one representative per orbit. In those treatments the symmetry is a property of a particular system and must be declared or detected. In ρ_G it is a property of the calculus.

Theorem 4.1 (Orbit quotient)**.** *Let $H \leq G$. Define $[P] = \{h \cdot P' : h \in H,\ P' \equiv P\}$ and $[P] \rightsquigarrow_H [Q]$ if $P \rightsquigarrow Q'$ for some $Q' \in [Q]$. Then $\rightsquigarrow_H$ is well-defined, and the quotient map $P \mapsto [P]$ is a functional bisimulation from $(\mathcal{P}, \rightsquigarrow)$ to $(\mathcal{P}/H, \rightsquigarrow_H)$. Consequently, for any set of states Φ closed under $\equiv$ and under the action of H, a state in Φ is reachable from P_0 if and only if an orbit in Φ/H is reachable from $[P_0]$.*

Proof. Well-definedness: if $h \cdot P \rightsquigarrow Q$ then $P \rightsquigarrow h^{-1} \cdot Q$ by Theorem 3.5, and $h^{-1} \cdot Q \in [Q]$. The bisimulation clauses are then immediate: every step of P projects to a step of $[P]$, and every step of $[P]$ is the projection of a step of P by the same argument. □

Three features distinguish this from the usual setting.

1. *Soundness for every subgroup.* Equivariance holds for the whole calculus, not for a given system, so the quotient is sound for any H and any initial process. Barbs move under the action, $h \cdot P \downarrow h \cdot x$ whenever $P \downarrow x$, so the invariant observations are orbit barbs $P \downarrow H{\cdot}x$, and for a spatial logic whose formulae mention names, with the action extended to formulae, satisfaction is equivariant: $g \cdot P \models g \cdot \varphi$ whenever $P \models \varphi$. Invariant formulae can be checked on the quotient.

2. *Effectiveness from the term.* The quotient pays when the reachable set is closed under H, which holds when $H \leq \mathrm{Stab}(P_0)$: then $P_0 \rightsquigarrow^* P$ gives $P_0 \equiv h \cdot P_0 \rightsquigarrow^* h \cdot P$. By Proposition 3.12 the largest such H is computed from P_0 with $|\mathrm{FN}(P_0)|$ checks; nothing is declared.

3. *A bound.* By Theorem 3.10, $|\mathrm{Stab}(P_0)|$ divides $|\mathrm{FN}(P_0)|$. The symmetry that can be extracted from the initial state is bounded by its number of free names, and the reduction factor on any orbit is at most $|H|$.

Canonicalisation, the choice of a representative of $[P]$, costs at most $|H|$ congruence-class comparisons for finite H, by taking the least of $\{h \cdot P\}_{h \in H}$ in any total order on normal forms. A spatial logic model checker for the rho calculus, such as the port of SLMC [2], is the natural home for this reduction, since spatial formulae already range over the structure that the action permutes.

5 Application II: symmetric rings

Rings of identical agents are the classical setting for symmetry in distributed computing. Orbit products give them a uniform syntax.

5.1 The construction

Let $r \in G$ have order n: a generator of $\mathbb{Z}/n$, or the rotation by $2\pi/n$ in $SO(2)$. Choose two anisotropic codes, for instance

$$a_f = @\langle \mathbf{0}, e \rangle !(\langle \mathbf{0}, e \rangle), \qquad a_t = \mathtt{for}(@\langle X, \delta \rangle \leftarrow @\langle \mathbf{0}, e \rangle)\mathbf{0},$$

and base names $f = @\langle a_f, e \rangle$ (a fork) and $t = @\langle a_t, e \rangle$ (a table, on which eating is announced). Write $f_k = r^k \cdot f$ and $t_k = r^k \cdot t$; indices are taken mod n. Since $r^k \cdot a_f = @\langle \mathbf{0}, r^k \rangle !(\langle \mathbf{0}, r^k \rangle)$, the codes of $f_0, \ldots, f_{n-1}$ are pairwise distinct, so the forks are n distinct channels and, by Proposition 3.8, every communication below is an ordinary one. Let

$$\begin{aligned} \mathit{Phil} &= \mathtt{for}(@\langle X, \delta \rangle \leftarrow f_0)\mathtt{for}(@\langle Y, \delta' \rangle \leftarrow f_1)\big(t_0!(\langle \mathbf{0}, e \rangle) \mid f_0!(\langle X, \delta \rangle) \mid f_1!(\langle Y, \delta' \rangle)\big), \\ \mathit{Fork} &= f_0!(\langle \mathbf{0}, e \rangle), \qquad R = \mathit{Phil} \mid \mathit{Fork}, \qquad \mathrm{Ring}_n = \textstyle\prod_{\langle r \rangle} R. \end{aligned}$$

The copy $r^k \cdot R$ is the k-th philosopher, who takes fork f_k and then fork f_{k+1}, announces on t_k, and returns both forks, together with fork f_k itself, carrying the token $\langle \mathbf{0}, r^k \rangle$. In the k-th philosopher the pattern is $@\langle [r^k]X, [r^k]\delta \rangle$. It still matches every token, and it binds what it receives in the philosopher's own frame: every philosopher runs literally the same body, in its own coordinates.

Proposition 5.1. *(i)* $\mathrm{Stab}(\mathrm{Ring}_n) = \langle r \rangle \cong \mathbb{Z}/n$ *when* $G = \mathbb{Z}/n$ *or* $G = SO(2)$.

(ii) Ring_n *reduces in* n *steps to the deadlocked state* $D_n = \prod_{\langle r \rangle} W$, *where* $W = \mathtt{for}(@\langle Y, \delta' \rangle \leftarrow f_1)\big(t_0!(\langle \mathbf{0}, e \rangle) \mid f_0!(\langle \mathbf{0}, e \rangle) \mid f_1!(\langle Y, \delta' \rangle)\big)$. *The state* D_n *has no reduct, has no barb on any* t_k, *and* $\mathrm{Stab}(D_n) = \langle r \rangle$.

(iii) For $n \geq 2$, Ring_n *also reaches states with a barb on some* t_k.

Proof. (i) The product is $\langle r\rangle$-invariant. Conversely $\mathrm{FN}(\mathrm{Ring}_n) = \{f_k, t_k\}_k$ has $2n$ elements. A symmetry maps fork names to fork names, since a send code is never congruent to a receive code, so by Proposition 3.12 it is $\nu(f_k)\nu(f_0)^{-1} = r^k$. (ii) For each k in turn, philosopher k receives the token of fork k; the coefficients are e; solving the pattern $@\langle [r^k]X, [r^k]\delta\rangle$ against $@\langle \mathbf{0}, r^k\rangle$ gives $\sigma = \{\mathbf{0}/X,\, e/\delta\}$, the same matcher for every k, and by Lemma 3.4 the continuation is $r^k \cdot W$. After n such steps no send on any f_k remains, so no (COMM) applies. Its symmetry group is computed as in (i). (iii) Let philosopher 0 take f_0 and then f_1 before philosopher 1 moves. □

The n steps to D_n are a single step and its $n-1$ rotations: the k-th is $r^k\cdot$ of the 0-th, placed in context. Equivariance is exactly the property that the adversary of Lehmann and Rabin exploits [6]: from a symmetric state, a scheduler that lets every copy take the corresponding step returns the system to a symmetric state, and a deterministic symmetric protocol cannot leave the symmetric states without help. Their theorem, that no deterministic, fully distributed and symmetric solution to the dining philosophers exists, can therefore be stated inside ρ_G with $\langle r\rangle$-invariance as the formal notion of symmetry. Its proof in the calculus is left to further work, together with the question of which ingredient of ρ_G is the symmetry breaker that their randomised solution requires. With summation read as an idempotent ensemble, as here, $\mathit{Phil}_L + \mathit{Phil}_R$ is demonic choice rather than a coin, and the deadlocked configuration remains in the ensemble.

Remark 5.2 (Isotropic rings). If the anisotropic code a_f is replaced by the isotropic code $\mathbf{0}$, the names $@\langle \mathbf{0}, r^k\rangle$ all share a code. Under the coded gate every orientation then communicates with every other, and a receive at r^j from a send at r^k transports the names of its own frame to r^k in one summand and to r^{2j-k} in the other. Under the K-gate with $K = \{e, r, r^{-1}\}$ only neighbours communicate, and the interaction topology is the Cayley graph of $\mathbb{Z}/n$. Rings of isotropic channels are therefore not rings of fixed agents but of agents that move as they communicate.

6 Application III: twisted roots

The note on radical fault tolerance [8] studies n-th roots of processes, R with $R^n \approx P$, where R^n is the n-fold parallel composition. Under a unique-decomposition hypothesis, which there concerns a behavioural equivalence, the existence of roots is governed by divisibility: every prime component of P must occur with multiplicity divisible by n, so that a component occurring once has no root. The group action offers a second way to divide a process into n equal parts: into n copies of one process at different orientations.

Definition 6.1. Let $H \le G$ be finite. A process R is an *H-twisted root* of P if $\prod_H R \equiv P$.

If $H \le \mathrm{Stab}(R)$ the twist is trivial, $\prod_H R \equiv R^{|H|}$, and an untwisted root is recovered. At the other extreme, if $f_k = r^k \cdot f$ as in Section 5, the process

$$P = f_0!(\langle \mathbf{0}, e\rangle) \mid f_1!(\langle \mathbf{0}, r\rangle) \mid \cdots \mid f_{n-1}!(\langle \mathbf{0}, r^{n-1}\rangle)$$

has every prime component with multiplicity one and so no n-th root, yet $f_0!(\langle \mathbf{0}, e\rangle)$ is a $\mathbb{Z}/n$-twisted root of it.

To say exactly when twisted roots exist we need unique decomposition. Twisted roots are defined up to structural congruence, and there, unlike up to bisimilarity, unique decomposition

is a lemma rather than a hypothesis. Call a closed process *flat* if neither $+$ nor $\mathbf{1}$ occurs in it outside prefixes and quotations. A *prime* is a process not congruent to $\mathbf{0}$ and not congruent to a composition of two processes neither of which is $\mathbf{0}$.

Lemma 6.2 (Unique decomposition up to $\equiv$)**.** *Every flat process is congruent to a parallel composition of receives and sends, unique up to $\equiv$ and permutation. The primes among flat processes are exactly the receives and the sends.*

Proof sketch. Consider the rewriting of Section 2: (CHAOS), (DIST) and idempotence oriented left to right, and (FRAME) used to make binding occurrences undecorated. It terminates, by a standard polynomial interpretation for the distributive and absorbing rules. It is confluent modulo AC; the critical pairs join, for instance $\mathbf{1} \mid (Q + R)$ rewrites to $\mathbf{1}$ directly and through $\mathbf{1} \mid Q + \mathbf{1} \mid R$, and $P \mid (Q + Q)$ rewrites to $P \mid Q$ through either idempotence or (DIST). Hence $P \equiv Q$ if and only if $\mathrm{nf}(P) =_{AC} \mathrm{nf}(Q)$. In a flat process no rule applies at top level, so $\mathrm{nf}(q_1 \mid \cdots \mid q_k) = \mathrm{nf}(q_1) \mid \cdots \mid \mathrm{nf}(q_k)$ with each q_i a receive or a send, and AC-equality of two such terms is equality of the multisets of their components. □

Behavioural unique decomposition, which is what the roots of [8] require, is a different and harder question; see Section 7. For a flat P write M_P for its multiset of primes. By Lemma 3.1 the action maps primes to primes, so H acts on primes and on such multisets, and each prime q has a stabiliser $\mathrm{Stab}_H(q) = \mathrm{Stab}(q) \cap H$.

Theorem 6.3 (Twisted divisibility)**.** *Let $H \leq G$ be finite and P flat. Then P has an H-twisted root if and only if*

(a) M_P is H-invariant, and

(b) for every prime q, $|\mathrm{Stab}_H(q)|$ divides the multiplicity $M_P(q)$.

Proof. ($\Rightarrow$) Let $\prod_H R \equiv P$. Then R is flat: a top-level sum of distinct configurations in $\mathrm{nf}(R)$ would survive into $\mathrm{nf}(\prod_H R)$, since parallel composition cancels in the free commutative monoid of Lemma 6.2, and a top-level $\mathbf{1}$ would make $P \equiv \mathbf{1}$. So by Lemma 6.2, $M_P = \sum_{h \in H} h \cdot M_R$, which is invariant. For a prime q,

$$M_P(q) = \sum_{h \in H} M_R(h^{-1} \cdot q) = |\mathrm{Stab}_H(q)| \sum_{q' \in H \cdot q} M_R(q'),$$

because each q' in the orbit of q equals $h^{-1} \cdot q$ for exactly $|\mathrm{Stab}_H(q)|$ elements h. ($\Leftarrow$) Choose one prime q from each H-orbit in M_P and let R be the composition of the chosen primes, each with multiplicity $M_P(q)/|\mathrm{Stab}_H(q)|$. By the same count, $\prod_H R$ gives each q' in the orbit of a chosen q multiplicity $|\mathrm{Stab}_H(q')| \cdot M_P(q)/|\mathrm{Stab}_H(q)| = M_P(q) = M_P(q')$, using that stabilisers along an orbit are conjugate and that M_P is invariant. □

Corollary 6.4. *Under the hypotheses of Theorem 6.3:*

(i) If H fixes every prime of P, then P has an H-twisted root if and only if $|H|$ divides every multiplicity. This is the root criterion of [8] with $n = |H|$.

(ii) If H acts freely on the primes of P, then P has an H-twisted root if and only if M_P is H-invariant: symmetry replaces divisibility.

(iii) By Theorem 3.10, $|\mathrm{Stab}_H(q)|$ divides $\gcd(|H|, |\mathrm{FN}(q)|)$. A prime whose number of free names is coprime to $|H|$ imposes no divisibility condition.

(iv) Chaos is its own twisted root for every H: $\prod_H \mathbf{1} \equiv \mathbf{1}$.

6.1 The stabiliser is the replication factor

Theorem 6.3 interpolates between two familiar ways of spreading a system over n sites. Take an orbit O of primes with stabiliser S. In $\prod_H R$, each prime of O is contributed by exactly $|S|$ of the copies.

- When $S = H$ every copy contributes the same component: this is *replication*, and the copies are those of an ordinary root.
- When S is trivial each component comes from exactly one copy, and the copies partition the system among themselves: this is *sharding*, in which every site runs the same program in its own frame.

In general the replication factor of a component is the order of its stabiliser, and by Corollary 6.4(iii) it divides both $|H|$ and its number of free names.

This bears directly on the arbitration problem of [8]. There, copies of a root contend for the same linear messages, and an arbiter must be installed at every contention site, at a cost of $\Theta(n)$ rendezvous per site. In a twisted root, identical primes, and hence identical channels, are shared only among the $|S|$ copies that replicate them. The contention that requires arbitration is confined to orbits with non-trivial stabiliser, and its width is $|S|$ rather than $|H|$.

Coupling. Copies with trivial stabilisers can still interact, through channels they share. Let

$$C(R) = \{\, h \in H \setminus \{e\} : \mathrm{code}(m) \equiv h \cdot \mathrm{code}(n) \text{ for some } m, n \in \mathrm{FN}(R) \,\}.$$

In the initial configuration, copies $k \cdot R$ and $k' \cdot R$ have a redex between them only if $k^{-1}k' \in C(R)$, so the static coupling between copies is the Cayley graph of H with connection set $C(R)$. For the ring of Section 5, $C(R) = \{r, r^{-1}\}$, since philosopher k shares fork f_{k+1} with philosopher $k+1$; R is a twisted root of Ring_n with trivial stabilisers, and its coupling graph is the cycle. When $C(R) = \emptyset$ the copies are disjoint in the initial configuration. Whether they stay disjoint depends on the names they can construct at run time, which is the namespace question that rho always raises and that the action does not settle.

Rotations again. With $G = SO(3)$ and H the icosahedral group, a twisted root spreads a system over 60 sites at the orientations of the icosahedron. By Corollary 6.4(iii), a component with a number of free names coprime to 60 is sharded, never replicated.

7 Further work

- *Behavioural twisted roots.* The structural theorem above should be lifted to $\prod_H R \approx P$ for an equivariant behavioural equivalence, relative to an observer as in [8], with orbit barbs as observations.
- *Arbitration in twisted roots.* The claim that arbitration is confined to orbits with non-trivial stabiliser should be made precise by combining the arbiter of [8] with the coupling graph.
- *Lehmann–Rabin in ρ_G.* State and prove the impossibility of deterministic symmetric solutions with $\langle r \rangle$-invariance as the notion of symmetry, and identify the symmetry breaker that a randomised solution needs. A probabilistic reading of summation is one candidate.

- *Transport on isotropic channels.* The behaviour of isotropic rings under the coded gate and under K-gates, where communication moves agents along a Cayley graph.
- *Symmetry reduction in practice.* Implement stabiliser computation and orbit canonicalisation in the spatial logic model checker, and measure the reduction on rings and twisted roots.
- *Behavioural unique decomposition.* Lemma 6.2 is structural. Up to bisimilarity, unique decomposition fails for the full rho calculus, since replication obtained by reflection is idempotent. The argument of Milner and Moller [12], in the generality of decomposition orders [10], needs a measure that adds across parallel composition and transitions that decompose. The phase-one translation into the rho combinators [11] supplies both: it is homomorphic on parallel composition and fully abstract relative to encoded contexts, and its weighted atom count adds across components. The candidate theorem is unique decomposition up to context-labelled bisimilarity for the fragment without drop in process position, proved in the target and transferred back, with divisor-closure of the image as the algebraic hinge. Corollary 3.7 then makes behavioural twisted roots statable.

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References

[1] M. A. Armstrong. *Groups and Symmetry*. Undergraduate Texts in Mathematics. Springer, 1988.

[2] L. Caires and L. Cardelli. A spatial logic for concurrency (part I). *Information and Computation*, 186(2):194–235, 2003.

[3] E. M. Clarke, E. A. Emerson, S. Jha, and A. P. Sistla. Symmetry reductions in model checking. In *CAV*, LNCS 1427, pages 147–158. Springer, 1998.

[4] E. A. Emerson and A. P. Sistla. Symmetry and model checking. *Formal Methods in System Design*, 9(1/2):105–131, 1996.

[5] C. N. Ip and D. L. Dill. Better verification through symmetry. *Formal Methods in System Design*, 9(1/2):41–75, 1996.

[6] D. Lehmann and M. O. Rabin. On the advantages of free choice: a symmetric and fully distributed solution to the dining philosophers problem. In *POPL*, pages 133–138. ACM, 1981.

[7] L. G. Meredith and M. Radestock. A reflective higher-order calculus. *Electronic Notes in Theoretical Computer Science*, 141(5):49–67, 2005.

[8] L. G. Meredith. Radical fault tolerance: roots of processes, arbitration, and the resolver that consensus supplies. F1R3FLY.io research note, 2026.

[9] L. G. Meredith. Where substitution happens. F1R3FLY.io research note, 2026.

[10] B. Luttik and V. van Oostrom. Decomposition orders: another generalisation of the fundamental theorem of arithmetic. *Theoretical Computer Science*, 335(2–3):147–186, 2005.

[11] F1R3FLY.io. Name-free combinators for the rho calculus, draft 3. F1R3FLY.io research note, 2026.

[12] R. Milner and F. Moller. Unique decomposition of processes. *Theoretical Computer Science*, 107(2):357–363, 1993.